

Madala Teja

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PROFESSIONAL SUMMARY

Software Development Engineer with 1 year of experience building scalable web applications, backend services, and data pipelines on cloud-native, distributed architectures. Proficient in Python, Java, Go, and C++, with a strong foundation in data structures, algorithms, and object-oriented design. Shipped production systems at Prodapt with measurable reliability and efficiency gains, owning broadly defined problems end-to-end in an agile environment. B.Tech in Computer Science and Engineering, 2025.

EDUCATION

Vasavi College of Engineering Hyderabad, India
B.Tech in Computer Science and Engineering 2021 – 2025

EXPERIENCE

- Software Development Engineer** Jul 2025 – Present
Prodapt Chennai, India
 - Architected an end-to-end data pipeline parsing large C++ codebases with Libclang, producing structured semantic context at scale and **improving system reliability by 30%**.
 - Built a distributed storage, index, and query layer on PostgreSQL and pgvector with hybrid retrieval (BM25 + cosine similarity + RRF), enabling low-latency, fault-tolerant lookups over large, high-throughput datasets.
 - Built internal web dashboards with React, giving engineering and operations teams real-time visibility into pipeline health and monitoring insights.
 - Integrated an automation and tool-execution layer into a Master Agent system via Model Context Protocol (MCP), replacing legacy Dialogflow triggers and expanding automation coverage.
 - Refactored a production codebase into reusable, modular components, reducing technical debt across teams in an agile SDLC.
- Software Engineer Intern** Mar 2025 – Jul 2025
Prodapt Chennai, India
 - Built an automated issue-categorization and ticket-routing tool, **cutting mean resolution time by 35%** for internal support teams.
 - Implemented data sanitization pipelines and dynamic knowledge-retrieval systems, improving contextual data quality by **40%** across downstream consumers.
- Software Developer Intern** Sep 2023 – Nov 2023
Technocolabs Hyderabad, India
 - Built CRUD REST APIs with Node.js, Spring Boot, MongoDB, and MySQL; implemented OAuth 2.0 and SSO authentication for access control and security compliance.

PROJECTS

- AI Workflow Automation System (Meeting Scheduling)** | *Python, MCP, REST APIs* Jun 2025 – Sep 2025
 - Built a multi-component automation system integrating Google Calendar and Gmail APIs with MCP orchestration, supporting event creation, updates, and reminders with error handling and audit logging.
- AI Incident Root Cause Analyzer** | *Python, Monitoring, RAG* Mar 2025 – May 2025
 - Built a log- and alert-processing pipeline that ingests operational data, extracts structured signals, and generates actionable root-cause insights, reducing mean debugging time.
- DDoS Detection System using Hybrid Deep Learning Models** | *Python, CNN, Transformer* Oct 2023 – Nov 2023
 - Designed a real-time detection framework combining CNN feature extraction with Transformer sequence modeling to reach **95% classification accuracy**.
- Canniecare** | *Python, Flask, PyTorch* Aug 2024 – Nov 2024
 - Developed an ML-backed web application for disease detection, achieving **92% model accuracy** and reducing diagnosis time by **40%**.

TECHNICAL SKILLS

Languages: Python, Java, Go, C++, TypeScript, SQL
Frontend & Web: React, TypeScript, JavaScript, HTML/CSS, Dashboards, Data Visualization
Backend: REST APIs, gRPC, Microservices, FastAPI, Flask, Node.js, Spring Boot, MongoDB, PostgreSQL
Distributed Systems: Data Pipelines, Event-Driven Architectures, Kafka, Fault Tolerance, Scalability
Cloud & Automation: AWS, GCP, Azure, Docker, Kubernetes, Automation Tooling (Build, Test, Monitoring)
Engineering: Data Structures & Algorithms, OOD, System Design, SDLC, Testing & Debugging, Agile
AI / ML: Machine Learning, Deep Learning, NLP, LLM Applications, RAG Pipelines, Agentic AI